

```
$ cat files/50-kptr-restrict.conf
kernel.kptr_restrict = 1

$ cat files/50-dmesg-restrict.conf
kernel.dmesg_restrict = 1

- name: Restrict access to kernel logs
  copy:
    src: ../../files/50-dmesg-restrict.conf
    dest: /etc/sysctl.d/50-dmesg-restrict.conf
    owner: root
    group: root
    mode: 0644

- name: Restrict access to kernel pointers in procs
  copy:
    src: ../../files/50-kptr-restrict.conf
    dest: /etc/sysctl.d/50-kptr-restrict.conf
    owner: root
    group: root
    mode: 0644
```

## Disable root login

The root SSH login should be disabled, by default. A remote user should only be able to log in to the system, and sudo if they have been given permissions. The same can be accomplished by setting *PermitRootlogin no* in */etc/ssh/sshd\_config* as shown below:

```
- name: Disable root login
  lineinfile:
    path: /etc/ssh/sshd_config
    regexp: '^#PermitRootLogin'
    line: 'PermitRootLogin no'
```

The console logins for root can also be disabled by commenting them out in */etc/securetty* file.

```
- name: Disable root console logins
  replace:
    path: /etc/securetty
    regexp: '^(?!#)(.*)'
    replace: '# \1'
```

## Execution

The entire playbook is provided below for reference:

```
---
- name: Harden Parabola
  hosts: parabola
  become: yes
  become_method: sudo

gather_facts: yes
tags: [security]

tasks:
  - name: Install pwgen
    package:
      name: pwgen
      state: latest

  - name: File permissions for /etc/shadow
    stat:
      path: /etc/shadow
      register: shadow

  - assert:
      that:
        - "shadow.stat.pw_name == 'root'"
        - "shadow.stat.readable == true"
        - "shadow.stat.rgrp == false"
        - "shadow.stat.roth == false"
        - "shadow.stat.writeable == true"
        - "shadow.stat.wgrp == false"
        - "shadow.stat.woth == false"
        - "shadow.stat.xgrp == false"
        - "shadow.stat.xoth == false"
        - "shadow.stat.xusr == false"

  - name: File permissions for /etc/passwd
    stat:
      path: /etc/passwd
      register: passwd

  - assert:
      that:
        - "passwd.stat.pw_name == 'root'"
        - "passwd.stat.readable == true"
        - "passwd.stat.rgrp == true"
        - "passwd.stat.roth == true"
        - "passwd.stat.writeable == true"
        - "passwd.stat.wgrp == false"
        - "passwd.stat.woth == false"
        - "passwd.stat.xgrp == false"
        - "passwd.stat.xoth == false"
        - "passwd.stat.xusr == false"

  - name: Update /etc/pam.d/passwd
    lineinfile:
      path: /etc/pam.d/passwd
      insertbefore: '^password.*required.*pam_unix.so'
      line: 'password          required          pam_cracklib.'
      so retry=2 minlen=10 difok=6 dcredit=-1 ucredit=-1 ocredit=-1
      lcredit=-1'

  - name: Check disk is encrypted
    shell: lsblk /dev/sda | grep / | grep crypt
```